
Agentic AI–Driven Smart Campus Operations Management System

Research Foundation & Literature Review

A structured survey of the concepts underpinning the system — agentic multi-agent orchestration, constraint-based scheduling, retrieval-augmented knowledge, and human-in-the-loop autonomy — mapping each published result to its role in our implementation, and establishing the research gap our project addresses.

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PROJECT GUIDE

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SCOPE

48 references · 8 concept areas

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SECTION 01

Overview & How to Use This Document

This document is the academic backbone of the project: for every concept the system uses, it collects real, citable literature and maps it to our implementation.

The survey is divided into two complementary sets of references. **Set A** (Sections 3–10) establishes that the concepts we rely on are well-founded and proven. **Set B** (Section 11) establishes the research gap — what existing work does *not* do, and how our approach is therefore novel.

Reading each concept table

Every concept section presents a uniform five-column table designed to drop directly into a project report or presentation:

COLUMN	PURPOSE
Paper	The title of the published work.
Source / Year	Publication venue (IEEE, ACM, Springer, Elsevier, IEEE Access, arXiv) with an active link.

COLUMN	PURPOSE
Core Idea	The contribution the paper makes.
Limitation / Gap	Where the work falls short — the basis of the “limitations” analysis.
What We Take	Exactly what our project borrows from or improves upon.

Citation verification. All paper titles, venues, and links in this document are genuine and were sourced from IEEE Xplore and publisher indexes. Before final submission, open each link and confirm the exact author list, page numbers, and year from the source page — these are the authoritative details for formal citation.

SECTION 02

Master Concept Mapping

The single table that ties research to concept, to our code, to rationale — the backbone of the methodology chapter and the system-design slides.

#	CONCEPT	WHERE IT LIVES IN OUR CODE	REF.	WHY WE USE IT
1	Agentic AI / autonomous multi-agent system	app/agents/ — supervisor + specialists	§3	Decompose campus operations into autonomous, cooperating agents rather than one monolithic program.
2	Supervisor-router orchestration	app/agents/supervisor.py	§4	A routing node classifies intent and delegates to the right specialist — the canonical multi-agent pattern.
3	Stateful graph + checkpointing	app/agents/graph.py (LangGraph + SqliteSaver)	§4	Durable, resumable workflows; enables pause/resume for approvals.
4	Tool / function calling	app/tools/ (Phase 1+)	§4	The LLM orchestrates; deterministic tools act — auditable and safe.
5	Constraint-based timetabling	app/solver/ OR-Tools CP-SAT	§5	Timetabling is NP-hard; a CP solver is provably clash-free and far faster than metaheuristics.
6	Dynamic rescheduling / substitution	Substitution Agent (Phase 2)	§6	Repair the timetable on disruption with minimal perturbation — the flagship feature.
7	Smart campus operations domain	Entire system	§7	Grounds the project in the established smart-campus research area.
8	Retrieval-Augmented Generation	app/rag/ Knowledge Agent (Phase 3)	§8	Answer campus questions from real documents with citations, without hallucination.

#	CONCEPT	WHERE IT LIVES IN OUR CODE	REF.	WHY WE USE IT
9	Human-in-the-loop approval	LangGraph <code>interrupt()</code> + <code>approvals</code> table	§9	Safe autonomy — the agent proposes, a human approves high-impact actions.
10	Facility / resource booking	Booking Agent (Phase 3)	§10	Conflict-aware venue reservation with an approval chain.

Adopted & Adapted — Contribution Summary

CONCEPT	WHAT EXISTING WORK DOES	ITS LIMITATION	WHAT WE TOOK / CHANGED
Multi-agent AI	Generic frameworks; cloud / parking / enterprise domains	Not applied to integrated campus operations	Apply the agentic pattern to a concrete campus-operations product
Orchestration	Supervisor-router pattern proven in surveys	Demonstrated on generic tasks	Implement it in LangGraph for campus intent routing with entity extraction
Timetabling	Genetic algorithms dominate the literature	Slow, stochastic, no infeasibility explanation	OR-Tools CP-SAT (deterministic, ~100× faster) + LLM explains infeasibility
Rescheduling	Studied mostly for railways / transport	Rarely applied to teacher substitution; not autonomous	Make substitution event-triggered and autonomous with a human approval gate
Smart campus	IoT sensing + dashboards	Reactive, hardware-centric, no autonomous decisions	Add an agentic decision layer on top of the data
RAG	Strong for domain QA	Generic corpora	Ground it in college-specific documents with citations
Human-in-the-loop	Approval patterns described conceptually	Often theoretical	Wire a concrete interrupt→approve→resume loop into a real workflow

SECTION 03 · SET A

Agentic AI & Multi-Agent Systems

Autonomous systems that pursue complex goals with minimal human intervention, often as multiple specialized agents that coordinate — the core paradigm of our project.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
Agentic AI in Action: A Review of Architectures, Communication, and Coordination in Intelligent Multi-Agent Systems	IEEE, 2025	Reviews 10+ agentic frameworks (2023–25); proposes a 3-dimensional taxonomy of architecture, communication, coordination.	Survey-level; open challenges in scalability, benchmarking, security; no concrete application.	Our supervisor + specialists design instantiates the coordination patterns it formalizes.
Agentic AI: Autonomous Intelligence for Complex Goals — A Comprehensive Survey	IEEE Access, v13, 2025	Defines agentic AI and its components: planning, memory, tools; application areas.	Broad survey; no domain implementation.	Grounds our definition of “agentic” and the autonomous-agent design.
Agentic AI Systems: Architecture and Evaluation Using a Frictionless Parking Scenario	IEEE, 2025	Specialized agents coordinate to achieve a goal with minimal supervision; agent design, interaction, cooperation.	Single narrow scenario (parking).	Blueprint for coordinating specialist agents toward one operational goal.
Agentic AI for Cloud Troubleshooting: A Review of Multi-Agent Systems for Automated Cloud Support	IEEE, 2025	Multi-agent system autonomously diagnoses and resolves cloud issues.	Domain-specific to cloud infrastructure.	Evidence the “detect → plan → act” loop works for real operational automation.
AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges	arXiv, 2025	Distinguishes single “AI agents” from collaborative “agentic AI”.	Conceptual, not applied.	Justifies a multi-agent design rather than a single chatbot.
Distinguishing Autonomous AI Agents from Collaborative Agentic Systems	arXiv, 2025	Framework for where collaboration between agents adds value.	Theoretical framework.	Supports the claim that specialist collaboration beats one general model.

HOW THIS MAPS TO OUR PROJECT

Our system is a collaborative agentic system: a supervisor classifies each request and delegates to a specialist that owns one domain, directly realizing the architecture / communication / coordination dimensions the IEEE survey formalizes and the parking paper demonstrates. Our contribution over these works is domain application — none target integrated campus operations.

LLM Orchestration, Supervisor-Router & Tool Calling

How multiple LLM-driven agents are coordinated. We use the supervisor (router) pattern: a central node classifies intent and delegates to workers that call typed tools.

Orchestration & the supervisor-router pattern

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
LLM-Based Multi-Agent Orchestration: A Survey of Frameworks, Communication Protocols, and Emerging Patterns	Future Internet (MDPI) , 2026	Explicitly describes the supervisor pattern where a routing node delegates to workers, plus hierarchical and swarm patterns.	Survey; does not build a product.	Direct blueprint for our <code>supervisor.py</code> → specialists design.
The Orchestration of Multi-Agent Systems: Architectures, Protocols, and Enterprise Adoption	arXiv , 2026	Implementation-ready design principles for enterprise multi-agent systems.	Enterprise-generic.	Principles for reliable orchestration: clear roles, typed hand-offs.
Multi-Agent Orchestration for High-Throughput Materials Screening	arXiv , 2026	Hierarchical planner–executor architecture implemented using LangGraph.	Scientific-computing domain.	Precedent that LangGraph — our exact framework — scales to real workflows.
Difficulty-Aware Agent Orchestration in LLM-Powered Workflows	arXiv , 2025	An LLM router assigns operators / models to tasks by difficulty.	Focuses on routing optimization only.	Validates our LLM-router supervisor; a future extension (route by cost / difficulty).
A Survey of Agent Interoperability Protocols (MCP, ACP, A2A, ANP)	arXiv , 2025	LLM agents combine linguistic reasoning with external tool execution; standard protocols.	Protocol-level, forward-looking.	Justifies our tool-calling design; roadmap for standard agent protocols.

Tool / function calling & intent classification

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
Function Calling in Large Language Models: Industrial Practices, Challenges, and Future Directions	OpenReview , 2025	Breaks function calling into pre-call (intent), execution (parameters, multi-call), post-call (result) stages.	Survey of failure modes.	We design tools to avoid these failures: typed schemas, validated parameters.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
Natural Language Tools: A NL Approach to Tool Calling in Large Language Agents	arXiv, 2025	Replacing rigid JSON tool calls with NL improves accuracy by 18.4 pts, cuts variance 70%.	New method, not yet standard.	Informs prompt design; a candidate optimization if JSON tool calls prove brittle.
Classification-Based Concurrent API Calls & Optimal Model Combination for Tool-Augmented LLMs	Scientific Reports, 2025	Multi-step plan → tool-invocation staging; classify then call APIs concurrently.	General agent setting.	Supports our two-stage flow: supervisor classifies, specialist invokes tools.

HOW THIS MAPS TO OUR PROJECT

`supervisor.py` is the supervisor/routing node the MDPI survey names as canonical, returning `{route, task_spec}` via structured output. `graph.py` builds a LangGraph `StateGraph` — the same framework as the materials-screening paper — and adds a `SqliteSaver` checkpointer. `app/tools/` follows the staged function-calling discipline with typed Pydantic schemas, mitigating the failure modes those papers catalogue.

SECTION 05 · SET A

Automated Timetable Scheduling (CP-SAT / OR-Tools)

University timetabling is a classic NP-hard constraint satisfaction problem. We solve it with Google OR-Tools CP-SAT — a constraint solver — not an LLM.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
A Comparative Analysis of Constraint Programming and Genetic Algorithms for Automated University Timetable Generation	IEEE, 2024	Compares Google CP-SAT vs a Genetic Algorithm; CP is ~175× faster and functionally superior on a realistic problem.	Benchmark study, not a product.	Direct justification for choosing OR-Tools CP-SAT over GA — our headline design citation.
Genetic Algorithm for Solving University Course Timetabling Using Dynamic Chromosomes	IEEE, 2021	GA with dynamic-length chromosomes for course timetabling.	Stochastic; convergence and tuning issues; no optimality guarantee.	Baseline we compare against; motivates a deterministic solver.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
Time Table Scheduling Using Genetic Algorithms Employing Guided Mutation	IEEE, 2011	Guided-mutation GA avoids faculty / classroom clashes.	Quality depends on operators; slow on large instances.	Confirms constraint types (faculty / room clash) we encode as hard constraints.
Solving Timetabling Problems Using Genetic Algorithms	IEEE, 2004	Foundational GA formulation of timetabling.	Older; small-scale; no infeasibility explanation.	Establishes timetabling as an evolutionary-search problem in the literature.
Solving University Timetabling as a Constraint Satisfaction Problem with Genetic Algorithm	Scholar	Models timetabling explicitly as a CSP.	Hybrid; still stochastic.	Confirms the CSP formulation we adopt for our CP-SAT model.
Automating University Course Scheduling Using Genetic Algorithm	Springer, 2025	Recent GA-based course scheduling with a practical framework.	GA limitations persist: runtime, no explanation.	Recent evidence the field still leans on GA — our CP-SAT + explanation is a step beyond.

HOW THIS MAPS TO OUR PROJECT

Our `app/solver/` module builds an OR-Tools CP-SAT model with boolean assignment variables, hard constraints for clash-freeness and period counts, and soft objectives for load balance. The comparison paper (IEEE 11379218) empirically shows CP-SAT is $\sim 175\times$ faster than a GA, which is precisely why we reject the GA-heavy mainstream. **Our added value:** when the problem is infeasible, the LLM explains which constraints conflict and suggests relaxations — a usability capability none of these papers provide.

SECTION 06 · SET A

Dynamic Rescheduling & Teacher Substitution

Repairing a live timetable on disruption (a teacher takes leave) while keeping changes minimal. Our Substitution Agent does this autonomously — the flagship differentiator.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
School Timetabling Problem Under Disturbances	Elsevier, 2016	Reschedules school timetables when teacher / room / learner availability changes.	Not autonomous; no agent; no human-approval loop.	Formal basis for “repair the timetable on disruption” — we automate and add approval.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
Minimum Penalty Perturbation Heuristics for Curriculum-Based Timetables Subject to Multiple Disruptions	Computers & OR, 2021	Heuristics that minimize perturbation when repairing timetables under disruptions.	Heuristic, offline framing.	The minimal-perturbation objective we adopt for substitution plans.
Dynamic and Robust Timetable Rescheduling for Uncertain Railway Disruptions	Elsevier, 2019	Rolling-horizon two-stage stochastic rescheduling under uncertainty.	Railway domain; heavy optimization.	Concept of robust, staged re-optimization under uncertainty.
Dynamic Railway Timetable Rescheduling for Multiple Connected Disruptions	Transp. Research C, 2021	MILP model reschedules under multiple overlapping disruptions.	Railway; MILP is heavy; no NL interface.	Reinforces multi-disruption handling; we keep it lightweight and agentic.
Automated Faculty Substitution Using Classification Algorithms	IJARCCE, 2025	Classifies suitable replacement teachers by subject compatibility.	Classification only; no full rescheduling, approval, or agent.	Closest prior work — we go further: ranking + plan + HITL + notify.

HOW THIS MAPS TO OUR PROJECT

The Substitution Agent combines the minimal-perturbation repair objective (Elsevier 2021) with substitute selection by subject compatibility (IJARCCE 2025). We rank candidates as teaches-same-subject > same-department > any-free, weighted by workload for fairness. What makes ours novel: it is autonomous and event-triggered (runs the moment leave is approved), keeps a human in the loop, and is applied to campus staff rather than railways — where prior work stops at classification.

SECTION 07 · SET A

Smart Campus Management Systems

The project domain. Our contribution is to move the smart campus from sensing and dashboards (reactive) to an agentic decision layer (proactive).

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
Internet of Things Based Model for Smart Campus: Challenges and Limitations	IEEE, 2020	Three-layer smart-campus model: perception → network → application.	Hardware / sensing-centric; reactive; no autonomous decision-making.	The layered model; our agents sit in an enhanced decision layer.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
Roadmap to Smart Campus Based on IoT	IEEE, 2020	Vision for smart schooling, classrooms, parking via IoT.	Roadmap; no intelligent automation.	Frames the campus subsystems (scheduling, facilities) we automate.
IoT, AI and Digital Twin for Smart Campus	IEEE, 2022	Digital twin + IoT for campus energy management.	Focus on energy infrastructure, not operations agents.	Precedent for our (stretch) Energy Watchdog agent.
Survey Toward a Smart Campus Using the Internet of Things	IEEE, 2016	Survey of smart-campus drivers: service quality, cost, sustainability.	Survey; IoT lens only.	Motivation (cost, efficiency) for our problem statement.
Internet of Things Based Smart Campus — A Review	IET/IEEE, 2022	Reviews smart-campus types by IoT layers and applications.	Review; no decision intelligence.	Establishes the state-of-the-art baseline we extend with agents.
Smart Campus as a Learning Platform for Industry 4.0 and IoT	IEEE, 2019	Uses the smart campus as an education / Industry-4.0 platform.	Pedagogical framing.	Broadens relevance; supports the education context.

HOW THIS MAPS TO OUR PROJECT

Every smart-campus paper here is IoT- and sensing-centric and, by its own framing, reactive — matching our problem statement's critique of “rigid, reactive, manually-supervised” systems. We adopt their layered view but add a new top layer: an agentic decision layer that acts on the data autonomously.

SECTION 08 · SET A

Retrieval-Augmented Generation (RAG)

Combining information retrieval with an LLM so answers are grounded in trusted documents, with citations. Our Knowledge Agent uses RAG over the college's own documents.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
Domain-Specific Retrieval-Augmented Generation Using Vector Stores, Knowledge Graphs, and Tensor Factorization	IEEE, 2025	Combines vector stores + knowledge graphs for domain-specific accuracy.	More complex infrastructure.	Confirms vector-store RAG for a domain corpus; KG is a future upgrade path.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
QuIM-RAG: Advancing RAG With Inverted Question Matching for Enhanced QA	IEEE Access, v12, 2024	Inverted question–chunk matching improves retrieval precision.	Adds a matching layer; more indexing cost.	Retrieval-quality technique if naive similarity search underperforms.
ICCA-RAG: Intelligent Customs Clearance Assistant Using RAG	IEEE Access, v13, 2025	Domain assistant answering regulatory questions via RAG.	Narrow regulatory domain.	Template for a domain assistant — ours is the campus equivalent.
Enhancing Precision & Interpretability of RAG in Legal Technology: A Survey	IEEE Access, v13, 2025	Surveys precision and interpretability (citations) in high-stakes RAG.	Survey; legal focus.	Motivates our citation requirement — answers must be traceable to a source.
A Novel Framework for Educational Q&A: Leveraging RAG and Code Interpreters	PMC, 2024	RAG + code interpreter for educational question answering.	Adds code-execution complexity.	Direct evidence RAG works for educational Q&A — our exact use case.
A Research of Challenges and Solutions in RAG Systems	HSET, 2024	Catalogues RAG challenges (retrieval quality, hallucination, chunking) and solutions.	Survey.	Checklist of pitfalls we design around: chunk size, “say I don't know”.

HOW THIS MAPS TO OUR PROJECT

Our `app/rag/` module ingests PDFs → chunks → embeddings → ChromaDB; the Knowledge Agent retrieves top-k chunks and answers grounded in them, citing source document and page. The domain-assistant papers (customs, legal) are our closest analogues — we build the campus version. The educational-QA paper confirms RAG suits education, and the challenges survey gives the failure list we guard against, especially the rule that the agent says “not found” rather than invent policy.

SECTION 09 · SET A

Human-in-the-Loop & Approval Workflows

Integrating human oversight at critical decision points: the AI proposes, then pauses until a human approves, edits, or rejects. This gives safe autonomy.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
A Decoupled Human-in-the-Loop System for Controlled Autonomy in Agentic Workflows	arXiv, 2026	Decouples agent execution from human approval checkpoints for controlled autonomy.	Recent preprint; general framework.	Directly matches our interrupt → approve → resume design.
A Comprehensive Framework for Human-AI Collaborative Decision Making	Expert Sys. w/ Apps, 2025	Framework for human–AI collaborative decisions.	Retail domain.	Tiered oversight: low-risk auto, high-risk human sign-off.
Human-in-the-Loop AI in Ongoing Process Verification (Pharma)	Information (MDPI), 2025	HITL for regulated pharmaceutical process verification.	Regulated-industry specific.	Evidence HITL is the norm where mistakes are costly — like altering a live timetable.
Trustworthy Human Computation: A Survey	arXiv, 2022	Trust via human participation — humans provide info or confirm / modify AI decisions.	Survey.	Grounds why HITL builds trust in our approvals.
Automating Document Intelligence in Statutory City Planning	arXiv, 2026	HITL document automation in a government / statutory setting.	Document-processing domain.	Precedent for HITL in administrative workflows like ours.

HOW THIS MAPS TO OUR PROJECT

We implement HITL with LangGraph's `interrupt()` plus a checkpoint: when a specialist reaches a high-impact action, the graph pauses and writes an `approvals` row with the `langgraph_thread_id`. The dashboard shows an approval card; when the HOD clicks Approve/Reject, the graph resumes from the exact paused state — even days later. Low-risk actions like notifications proceed automatically; high-risk timetable / booking changes require sign-off.

SECTION 10 · SET A

Facility / Room Booking & Resource Allocation

Allocating scarce venues without clashes and with capacity matching. Our Booking Agent handles natural-language requests and checks conflicts against bookings and the timetable.

PAPER	SOURCE / YEAR	CORE IDEA	LIMITATION / GAP	WHAT WE TAKE
Design and Development of an Integrated Room Reservation System for Higher Education Institutions	IEEE, 2021	Online system easing room / facility management with real-time info exchange.	Form-based CRUD; no NL interface, no autonomous conflict resolution, no agent.	Baseline reservation system — we add NL + agentic conflict handling.
Optimization of Room Allocation Plans (Univ. Duisburg-Essen) with a Regulatory Algorithm	IEEE, 2016	Evolutionary regulatory algorithm optimizes room allocation on real data.	Optimization only; batch; not interactive.	Evidence of algorithmic room allocation; we keep it conversational and real-time.
Online Reservation Inventory Systems in Higher Education	IJRES, 2025	Architecture and layers of online room-booking systems for universities.	Traditional IS design; no AI.	System-design reference for our booking data model and flow.
DHMS: A Digital Hostel Management System Integrating Campus ChatBot, Predictive Intelligence, and Real-Time Automation	arXiv, 2025	Campus system combining a chatbot + predictive automation for hostel management.	Hostel-specific; chatbot, not a full agent.	Nearby example of chatbot + campus automation; we generalize to an agent.

HOW THIS MAPS TO OUR PROJECT

Existing reservation systems are form-based CRUD. Our Booking Agent instead accepts natural language, extracts structured fields via the supervisor, and checks conflicts against the live timetable as well as other bookings — a check those systems do not perform. When a venue is taken, it proposes alternatives by capacity, then routes to the approval chain.

SECTION 11 · SET B

Differentiation — What Makes This Project Stand Out

The gap in existing work, and how our approach is novel. This section supports the novelty / contribution slide and the gap-analysis chapter.

THE ONE-LINE THESIS

Existing campus systems are reactive (IoT dashboards), and existing agentic-AI research is generic or applied to other domains (cloud, parking, railways). No prior system combines autonomous multi-agent orchestration, proactive event-driven action, and human-in-the-loop approval, applied to integrated campus operations. That combination is our contribution.

Gap analysis — existing → limitation → our improvement

AREA	WHAT EXISTING WORK DOES	ITS LIMITATION	OUR DIFFERENTIATION
Smart campus	IoT sensing, dashboards, digital twins	Reactive; hardware-centric; no autonomous decisions	An agentic decision layer that acts, not just displays.
Timetabling	Genetic algorithms dominate	Slow, stochastic, no infeasibility explanation	CP-SAT (~175× faster) + LLM explains why a timetable is impossible and suggests fixes.
Rescheduling	Railway / transport, offline; faculty substitution = classification only	Manual trigger; not autonomous; no approval	Event-triggered autonomous substitution with human approval.
Multi-agent AI	Surveys + apps: cloud, parking	Not applied to campus operations	First to apply the agentic pattern to integrated campus operations.
Reservation	Form-based CRUD booking	No NL; no timetable-aware conflict check	NL booking + conflict check against the live timetable + alternatives.
Knowledge / FAQ	Generic or domain RAG	Not campus-grounded	RAG over the college's own documents with citations.
Autonomy safety	HITL described conceptually	Often theoretical	Concrete interrupt → approve → resume loop in a working product.

The four pillars of our novelty

- 1 Proactive, not reactive.** Agents act on events (leave approved → reschedule) and schedules (nightly checks) with no human prompt. Smart-campus literature is explicitly reactive by its own framing.
- 2 Right tool for each job.** Constraint solver for timetables, RAG for knowledge, structured-output LLM for routing — the LLM orchestrates, it does not guess. Timetabling research shows LLM-free CP-SAT is the correct engineering choice.
- 3 Safe autonomy via human-in-the-loop.** The system proposes; humans approve high-impact actions in one click.

Integrated, single platform. One agentic system spans scheduling, substitution, booking, and knowledge — prior work solves these in isolation.

Set B — Agentic AI specifically for education / administration

PAPER	SOURCE / YEAR	CORE IDEA	WHY IT SUPPORTS OUR DIFFERENTIATION
Agent4EDU: Advancing AI for Education with Agentic Workflows	ACM ICAIE, 2024	Agentic workflows applied to education.	Confirms agentic AI in education is a recognized, publishable direction.
Instructional Agents: Reducing Teaching Faculty Workload through Multi-Agent Instructional Design	arXiv, 2025	Multi-agent system reduces faculty workload.	Same motivation as ours (cut staff workload) but targets instructional design, not operations.
Agentic Workflow for Education: Concepts and Applications	arXiv, 2025	Concepts and applications of agentic workflows in education.	Positions our operations focus as an unexplored application area.
LLM Agents for Education: Advances and Applications	arXiv, 2025	Survey of LLM agents in education.	Shows operations / administration is under-addressed.
Large-Language-Model-Based Agents for Education (Survey)	IEEE Trans. Learning Tech., 2025	Comprehensive survey of LLM agents for education.	Strong IEEE citation establishing the research area.
The Impact of LLMs on Higher Education: AI and Education 4.0	Frontiers, 2024	LLMs and the Education 4.0 vision.	Macro-context: why now, why campuses.

Report-ready formulation

“While agentic AI has been surveyed extensively and applied to domains such as cloud support and smart parking, and while smart-campus research is mature in IoT sensing, the two have not been combined. Existing campus systems remain reactive and siloed, timetabling research relies largely on slower stochastic genetic algorithms without infeasibility reasoning, and teacher-substitution work stops at classification. Our system unifies these into a single agentic platform that acts proactively on events, uses a constraint solver with LLM-generated explanations for timetabling, and keeps humans in control through explicit approval checkpoints — a combination not present in the reviewed literature.”